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GATE ECE · NETWORK THEORY · CHAPTER-WISE PYQS

Transient Response

GATE ECE Network Theory · Previous Year Questions with Answer Key

56	35	1987 – 2024	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	17 questions
Marks 2	29 questions
Marks 3	1 question
Marks 5	7 questions
Marks 8	1 question
Marks 10	1 question
Answer key	at the end

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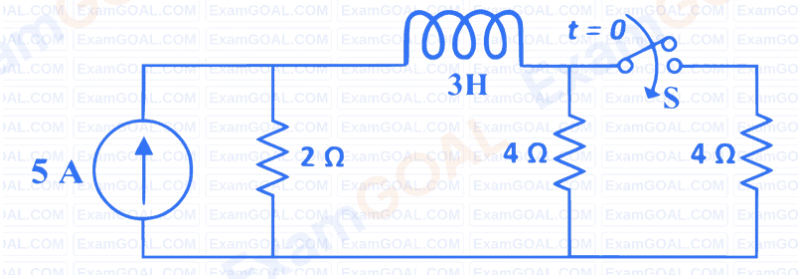
1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

Marks 1 17 questions

Q1. GATE ECE 2024

In the circuit given below, the switch S was kept open for a sufficiently long time and is closed at time $t = 0$. The time constant (in seconds) of the circuit for $t > 0$ is _____ .

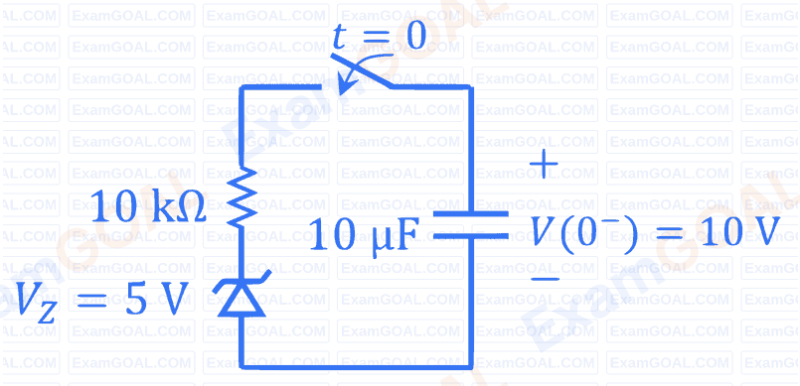
Open solution



Q2. GATE ECE 2024

As shown in the circuit, the initial voltage across the capacitor is 10 V, with the switch being open. The switch is then closed at $t = 0$. The total energy dissipated in the ideal Zener diode $(V_Z = 5V)$ after the switch is closed (in mJ, rounded off to three decimal places) is _____.

Open solution

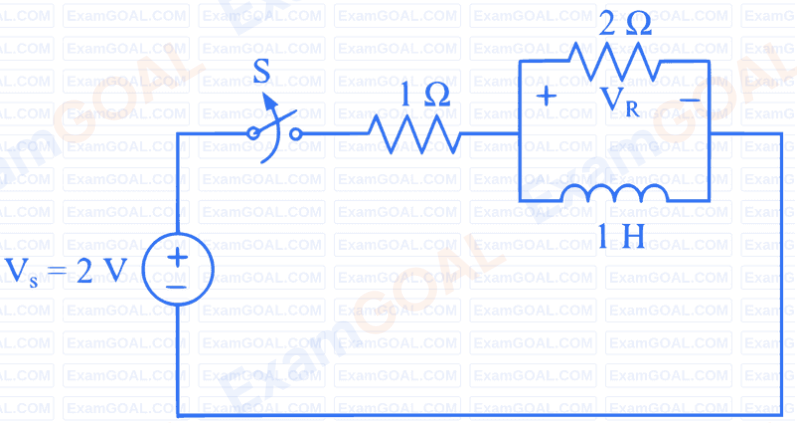


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Q3. GATE ECE 2023

In the circuit shown below, switch S was closed for a long time. If the switch is opened at t = 0, the maximum magnitude of the voltage V_R, in volts, is _____ (rounded off to the nearest integer).

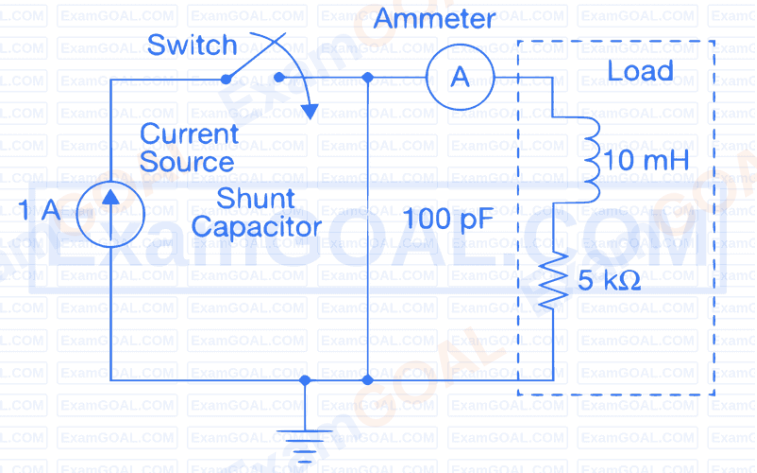
Open solution



Q4. GATE ECE 2021

The circuit in the figure contains a current source driving a load having an inductor and a resistor in series, with a shunt capacitor across the load. The ammeter is assumed to have zero resistance. The switch is closed at time $t = 0$.

Open solution



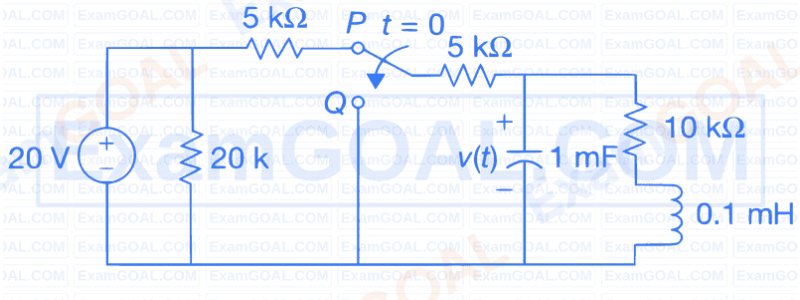
Initially, when the switch is open, the capacitor is discharged, and the ammeter reads zero ampere. After the switch is closed, the ammeter reading keeps fluctuating for some time till it settles to a final steady value. The maximum ammeter reading that one will observe after the switch is closed (rounded off to two decimal places) is ____ A.

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Q5. GATE ECE 2021

Open solution

The switch in the circuit in the figure is in position P for a long time and then moved to position Q at time $t = 0$



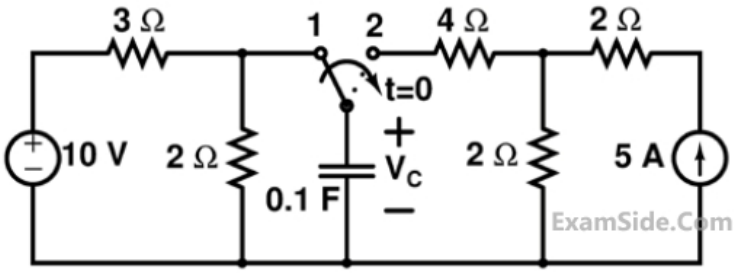
The value of $\frac{dv(t)}{dt}$ at $t = 0^+$is

(A) $3\ \mathrm{V/s}$ (B) $-5\ \mathrm{V/s}$ (C) $0\ \mathrm{V/s}$ (D) $-3\ \mathrm{V/s}$

Q6. GATE ECE 2016 Set 2

Open solution

The switch has been in position 1for a long time and abruptly changes to position 2 at t = 0.



If time t is in seconds, the capacitor voltage V_C (in volts) for t>0 is given by

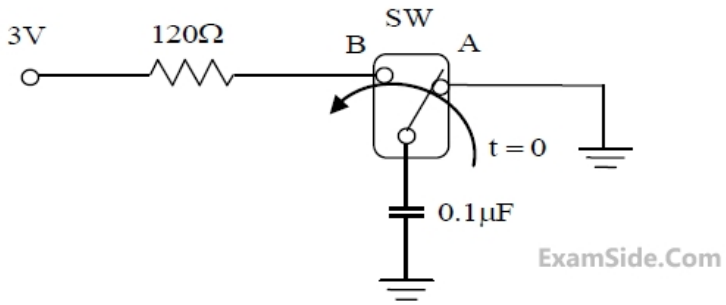
(A) 4(1 - exp(-t/0.5)) (B) 10 - 6exp(-t/0.5)

(C) 4(1 - exp(-t/0.6)) (D) 10 - 6exp(-t/0.6)

Q7. GATE ECE 2015 Set 1

Open solution

In the circuit shown, the switch SW is thrown from position A to position B at time t = 0. The energy (in μJ) taken from the 3V source to charge the 0.1μF capacitor from 0V to 3V is



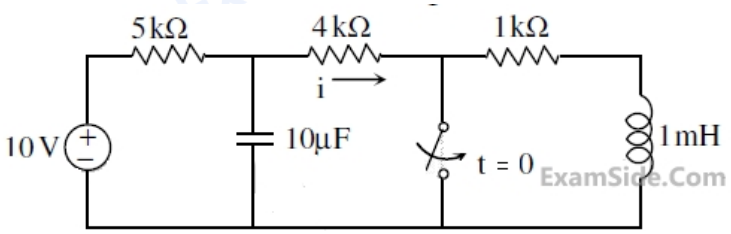
(A) 0.3 (B) 0.45 (C) 0.9 (D) 3

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Q8. GATE ECE 2014 Set 2

Open solution

In the figure shown, the ideal switch has been open for a long time. If it is closed at t=0, then the magnitude of the current (in mA) through the $4k\Omega$ resistor at t = 0^+ is _____.



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Q9. GATE ECE 2014 Set 3

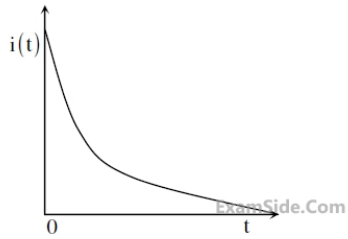
Open solution

A series RC circuit is connected to a DC voltage source at time t = 0. The relation between the source voltage V_S, the resistance R, the capacitance C, and the current i(t) is given below:

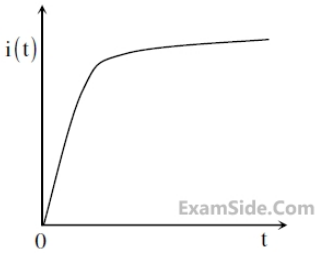
$$V_s = Ri(t) + \frac{1}{c}\int_0^t i(u)du$$

Which one of the following represents the current f(t)?

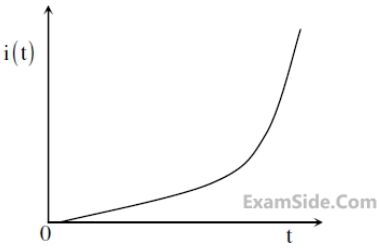
(A)



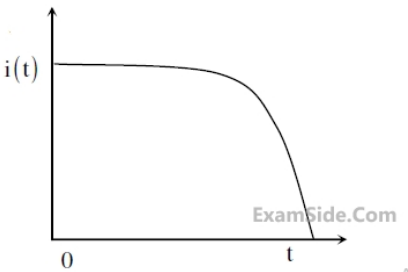
(B)



(C)



(D)

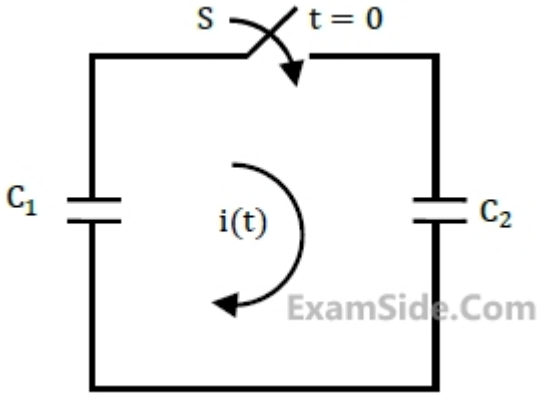


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Q10. GATE ECE 2012

In the following figure, C_1 and C_2 are ideal capacitors. C_1 has been charged to 12 V before the ideal switch S is closed at t = 0. The current i(t) for all t is.

Open solution



(A) zero
(B) a step function
(C) an exponentially decaying function
(D) an impulse function

Q11. GATE ECE 2011

If the unit step response of a network is $(1 - e^{-\alpha t})$, then its unit impulse response is

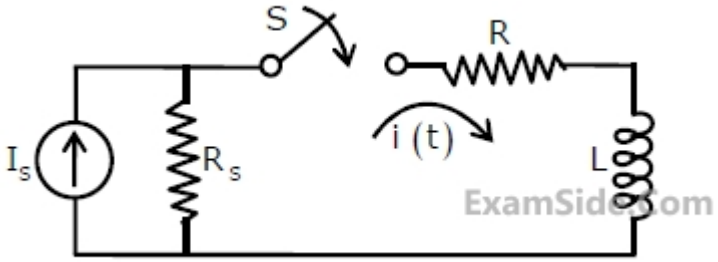
(A) $\alpha\, e^{-\alpha t}$ (B) $\alpha^{-1}\, e^{-\alpha t}$ (C) $(1 - \alpha^{-1})e^{-\alpha t}$ (D) $(1 - \alpha)e^{-\alpha t}$

Open solution

Q12. GATE ECE 2008

In the following circuit, the switch S is closed at t=0. The rate of change of current $\frac{di}{dt}(0^+)$ is given by

Open solution



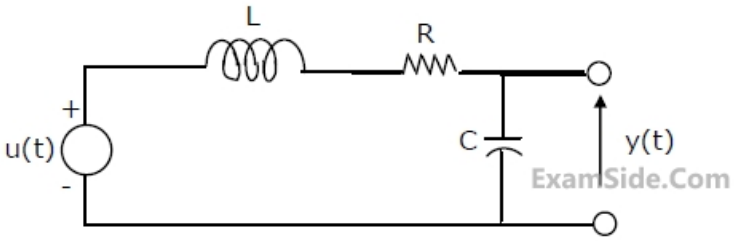
(A) 0 (B) $\frac{R_s I_s}{L}$ (C) $\frac{(R+R_s)I_s}{L}$ (D) ∞

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Q13. GATE ECE 2005

The condition on R, L and C such that the step response y(t) in figure has no oscillations, is

Open solution



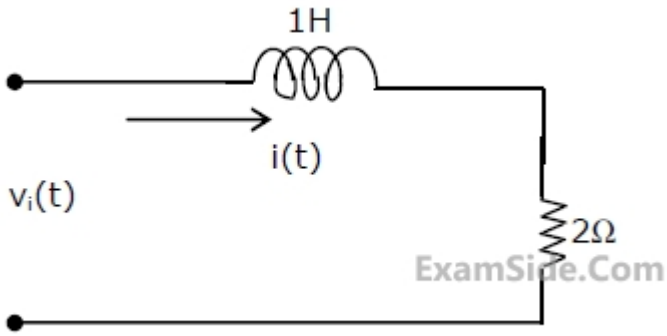
(A) $R \geq \frac{1}{2}\sqrt{\frac{L}{C}}$ (B) $R \geq \sqrt{\frac{L}{C}}$ (C) $R \geq 2\sqrt{\frac{L}{C}}$ (D) $R \geq \frac{1}{\sqrt{LC}}$

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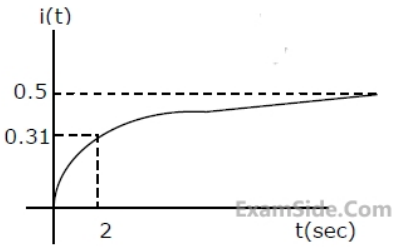
Q14. GATE ECE 2004

Open solution

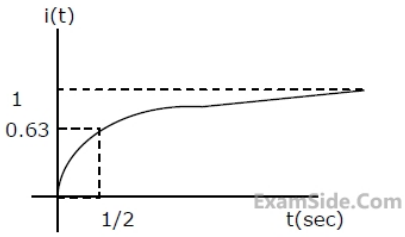
For the R-L circuit shown in Fig, the input voltage. $v_i(t) = u(t)$ The current i(t) is



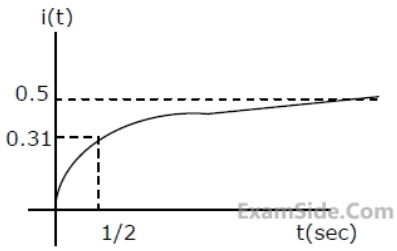
(A)



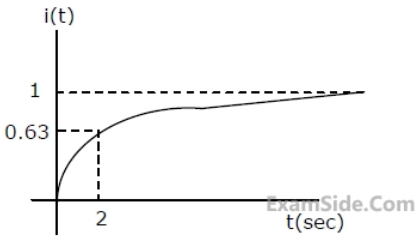
(B)



(C)



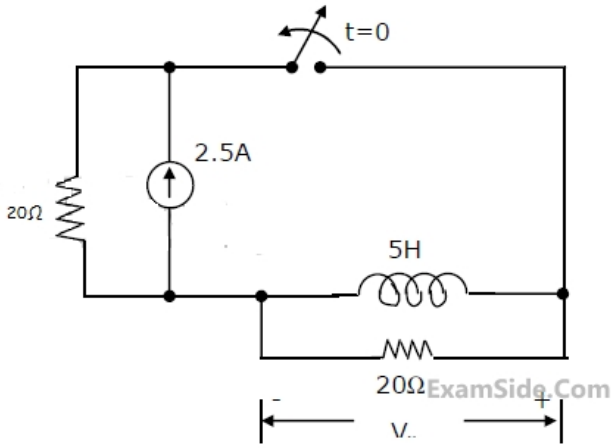
(D)



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Q15. GATE ECE 2002

In figure, the switch was closed for a long time before opening at t = 0. the voltage V_x at t = 0^+ is



Open solution

(A) 25 V (B) 50 V (C) -50 V (D) 0 V

Q16. GATE ECE 1994

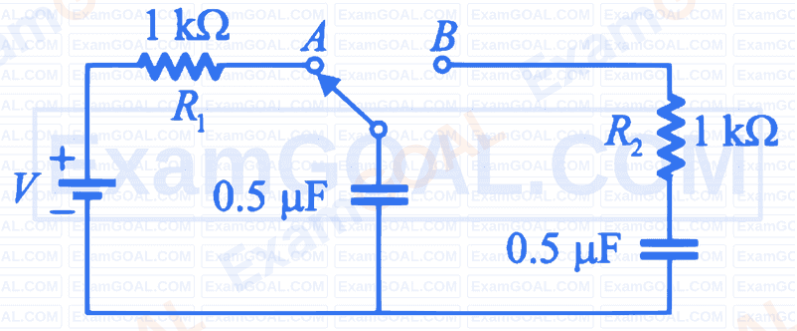
A ramp voltage, v(t) = 100 t Volts, is applied to an RC differentiating circuit with R = $5k\Omega$ and C = $4\mu F$.The maximum output voltage is

(A) 0.2 volt (B) 2.0 volts (C) 10.0 volts (D) 50.0 volts

Open solution

Q17. GATE ECE 1992

For the circuit shown below different time constants are given :



1. 1. 0.5×10^{-3}sec
2. 2. 2×10^{-3}sec
3. 3. 0.25×10^{-3}sec
4. 4. 10^{-3}sec

What are the charging and discharging times respectively :

(A) 1, 2 (B) 2, 3 (C) 1, 3 (D) 2, 4

Open solution

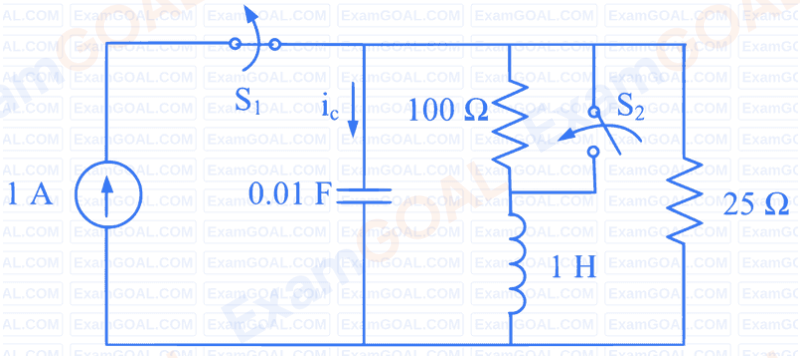
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Marks 2 29 questions

Q18. GATE ECE 2023

Open solution

The switch S_1 was closed and S_2 was open for a long time. At t = 0, switch S_1 is opened and S_2 is closed, simultaneously. The value of $i_c(0^+)$, in amperes, is

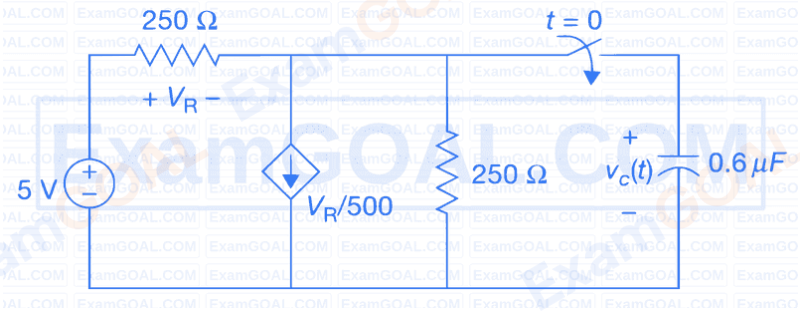


(A) 1 (B) -1 (C) 0.2 (D) 0.8

Q19. GATE ECE 2021

Open solution

In the circuit shown in the figure, the switch is closed at time $t = 0$, while the capacitor is initially charges to -5 V (i.e., $(\theta_C(0) = -5\,\text{V})$.



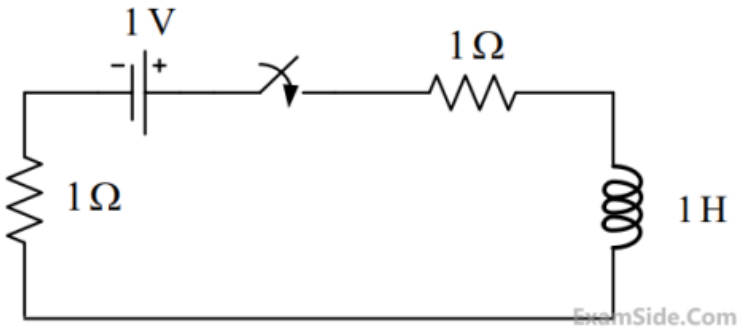
The time after which the voltage across the capacitor becomes zero (rounded off to three decimal places) is ____ ms .

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Q20. GATE ECE 2018

Open solution

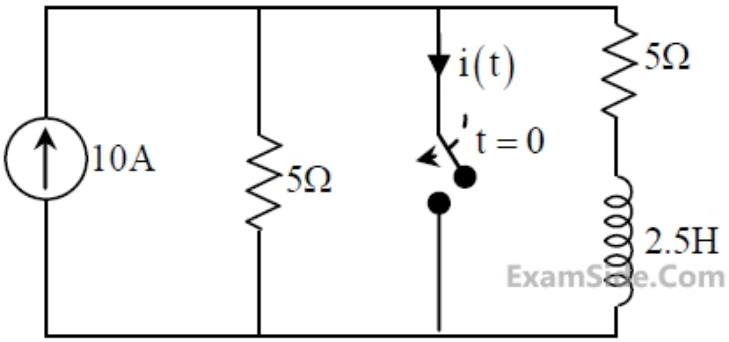
For the circuit given in the figure, the magnitude of the loop current (in amperes, correct to three decimal places) 0.5 second after closing the switch is _____.



Q21. GATE ECE 2017 Set 2

Open solution

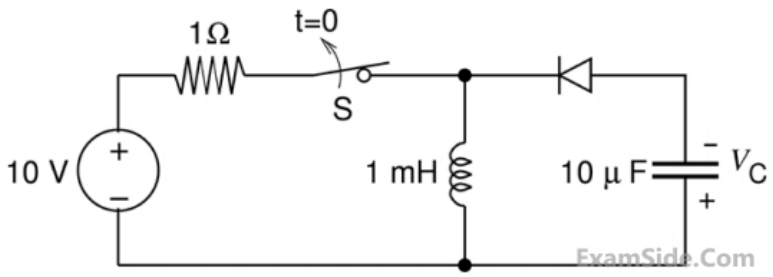
The switch in the circuit, shown in the figure, was open for a long time and is closed at t = 0. The current i(t) (in ampere) at t = 0.5 seconds is _____



Q22. GATE ECE 2016 Set 2

Open solution

The switch S in the circuit shown has been closed for a long time. It is opened at time t = 0 and remains open after that. Assume that the diode has zero reverse current and zero forward voltage drop.



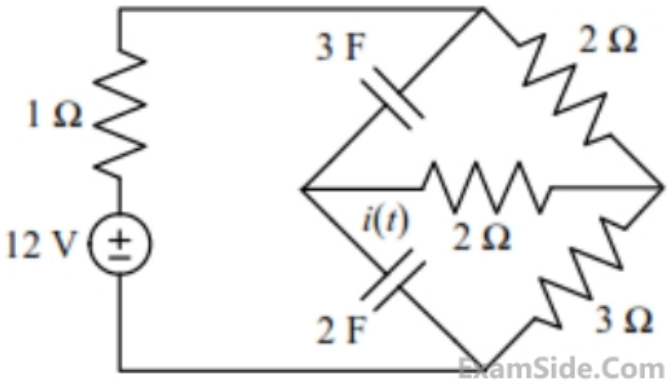
The steady state magnitude of the capacitor voltage V_C (in volts) is _____.

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Q23. GATE ECE 2016 Set 3

Open solution

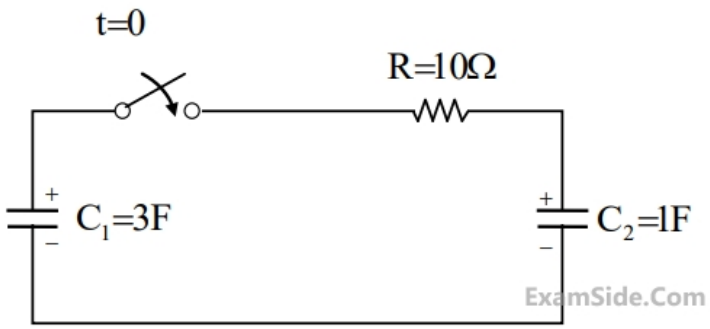
Assume that the circuit in the figure has reached the steady state before time t = 0 when the 3 Ω resistor suddenly burns out, resulting in an open circuit. The current i(t) (in ampere) at $t = 0^+$ is _____.



Q24. GATE ECE 2015 Set 2

Open solution

In the circuit shown, the initial voltages across the capacitors C_1 and C_2 are 1 V and 3 V, respectively. The switch is closed at time t = 0. The total energy dissipated (in Joules) in the resistor R until steady state is reached is _____.

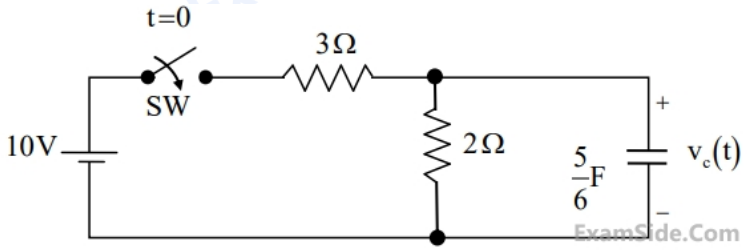


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Q25. GATE ECE 2015 Set 1

Open solution

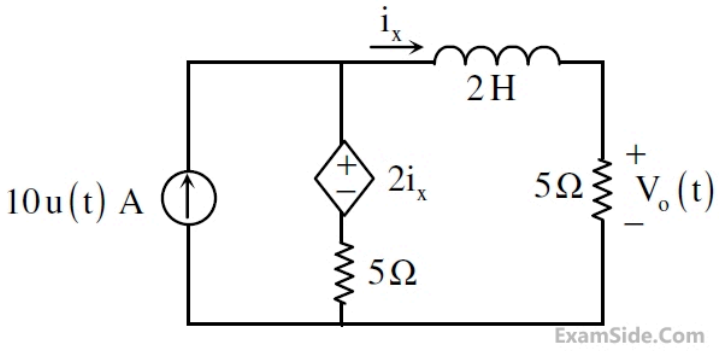
In the circuit shown, switch SW is closed at t = 0. Assuming zero initial conditions, the value of $v_c(t)$ (in Volts) at t= 1 sec is _____.



Q26. GATE ECE 2014 Set 4

Open solution

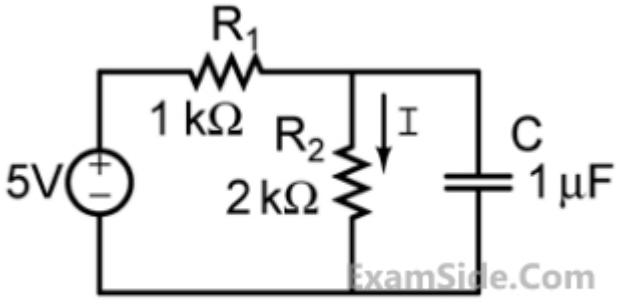
In the circuit shown in the figure, the value of $V_0(t)$ (in Volts) for $t \to \infty$ is _____.



Q27. GATE ECE 2014 Set 2

Open solution

In the figure shown, the capacitor is initially uncharged. Which one of the following expressions describes the current I(t) (in mA) for t > 0?



(A) $I(t) = \frac{5}{3}\left(1 - e^{\frac{-t}{\tau}}\right), \tau = \frac{2}{3}\ m\ sec$

(B) $I(t) = \frac{5}{2}\left(1 - e^{\frac{-t}{\tau}}\right), \tau = \frac{2}{3}\ m\ sec$

(C) $I(t) = \frac{5}{3}\left(1 - e^{\frac{-t}{\tau}}\right), \tau = 3\ m\ sec$

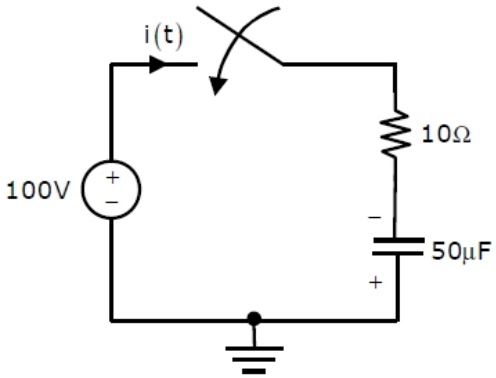
(D) $I(t) = \frac{5}{2}\left(1 - e^{\frac{-t}{\tau}}\right), \tau = 3\ m\ sec$

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Q28. GATE ECE 2011

Open solution

In the circuit shown below, the initial charge on the capacitor is 2.5 mC, with the voltage polarity as indicated. The switch is closed at time t=0. The current i(t) at a time t after the switch is closed is



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(A) $i(t) = 15 \exp(-2 \times 10^3 t)$ A

(B) $i(t) = 5 \exp(-2 \times 10^3 t)$ A

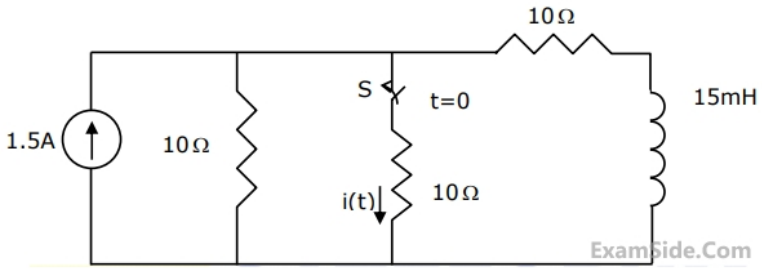
(C) $i(t) = 10 \exp(-2 \times 10^3 t)$ A

(D) $i(t) = -5 \exp(-2 \times 10^3 t)$ A

Q29. GATE ECE 2010

Open solution

In the circuit shown, the switch S is open for a long time and is closed at t=0. The current i(t) for $t \geq 0^+$ is



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(A) $i(t) = 0.5 - 0.125e^{-1000t}$ A

(B) $i(t) = 1.5 - 0.125e^{-1000t}$ A

(C) $i(t) = 0.5 - 0.5e^{-1000t}$ A

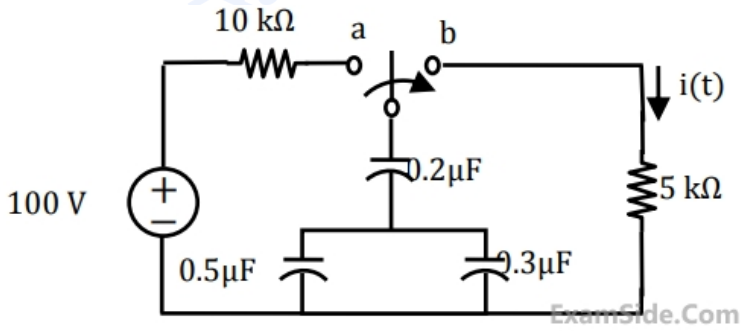
(D) $i(t) = 0.375e^{-1000t}$ A

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Q30. GATE ECE 2009

Open solution

The switch in the circuit shown was on position 'a' for a long time and is moved to position 'b' at time t = 0. The current i(t) for t > 0 is given by



(A) $0.2e^{-125t}$ u(t) mA

(B) $20e^{-1250t}$ u(t) mA

(C) $0.2e^{-1250t}$ u(t) mA

(D) $20e^{-1000t}$ u(t) mA

Q31. GATE ECE 2009

Open solution

The time domain behavior of an RL circuit is represented by

$$L\frac{di(t)}{dt} + Ri = V_0\left(1 + Be^{-Rt/L}\sin t\right)u(t)$$

For an initial current of i(0) = $\frac{V_0}{R}$, the steady state value of the current is given by

(A) $i(t) \to \frac{V_0}{R}$

(B) $i(t) \to \frac{2V_0}{R}$

(C) $i(t) \to \frac{V_0}{R}(1+B)$

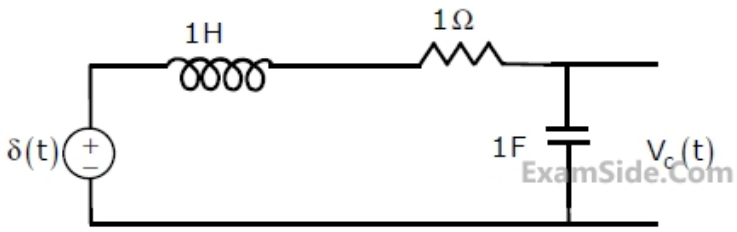
(D) $i(t) \to \frac{2V_0}{R}(1+B)$

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Q32. GATE ECE 2008

The following series RLC circuit with zero initial conditions is excited by a unit impulse function $\delta(t)$.

Open solution



For t > 0, the output voltage $V_c(t)$ is

(A) $\frac{2}{\sqrt{3}}\left(e^{-\frac{1}{2}t} - e^{-\frac{\sqrt{3}}{2}t}\right)$

(B) $\frac{2}{\sqrt{3}}te^{-\frac{1}{2}t}$

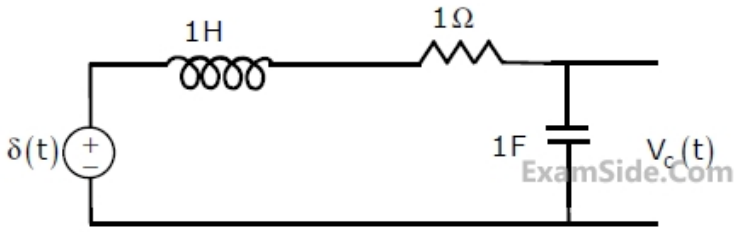
(C) $\frac{2}{\sqrt{3}}e^{-\frac{1}{2}t}\cos\left(\frac{\sqrt{3}}{2}t\right)$

(D) $\frac{2}{\sqrt{3}}e^{-\frac{1}{2}t}\sin\left(\frac{\sqrt{3}}{2}t\right)$

Q33. GATE ECE 2008

The following series RLC circuit with zero initial conditions is excited by a unit impulse function $\delta(t)$.

Open solution



For t > 0, the voltage across the resistor is

(A) $\frac{1}{\sqrt{3}}\left(e^{-\frac{\sqrt{3}}{2}t} - e^{-\frac{1}{2}t}\right)$

(B) $e^{-\frac{1}{2}t}\left[\cos\left(\frac{\sqrt{3}}{2}t\right) - \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}t\right)\right]$

(C) $\frac{2}{\sqrt{3}}e^{-\frac{1}{2}t}\sin\left(\frac{\sqrt{3}}{2}t\right)$

(D) $\frac{2}{\sqrt{3}}e^{-\frac{1}{2}t}\cos\left(\frac{\sqrt{3}}{2}t\right)$

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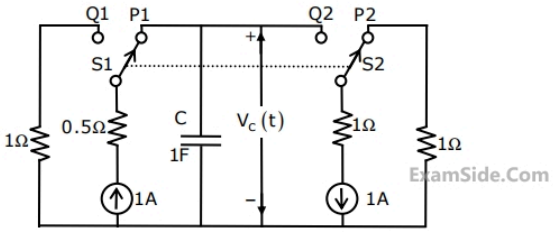
Q34. GATE ECE 2008

Open solution

The circuit shown in the figure is used to charge the capacitor C alternately from two current sources as indicated. The switches S1 and S2 are mechanically coupled and connected as follows

For $2nT \le t < (2n+1)T$, $(n = 0, 1, 2, \ldots)$ S1 to P1 and S2 to P2

For $(2n+1)T \le t < (2n+2)T$, $(n = 0, 1, 2, \ldots)$ S1 to Q1 and S2 to Q2



Assume that the capacitor has zero initial charge. Given that u(t) is a unit step function, the voltage $V_C(t)$ across the capacitor is given by

(A) $\sum_{n=0}^{\infty} (-1)^n \, tu(t - nT)$

(B) $u(t) + 2\sum_{n=0}^{\infty} (-1)^n \, u(t - nT)$

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(C) $tu(t) + 2\sum_{n=0}^{\infty} (-1)^n \, (t - nT)u(t - nT)$

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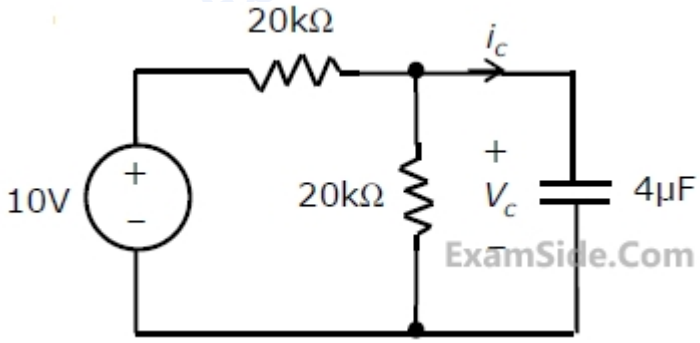
(D) $\sum_{n=0}^{\infty} \left[0.5 - e^{-(t-2nT)} + 0.5e^{-(t-2nT-T)}\right]$

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Q35. GATE ECE 2007

Open solution

In the circuit shown, V_C is 0 volts at t = 0 sec. For t > 0, the capacitor current $i_C(t)$, where t is in seconds, is given by



(A) 0.50 exp(-25t) mA

(B) 0.25 exp(-25t) mA

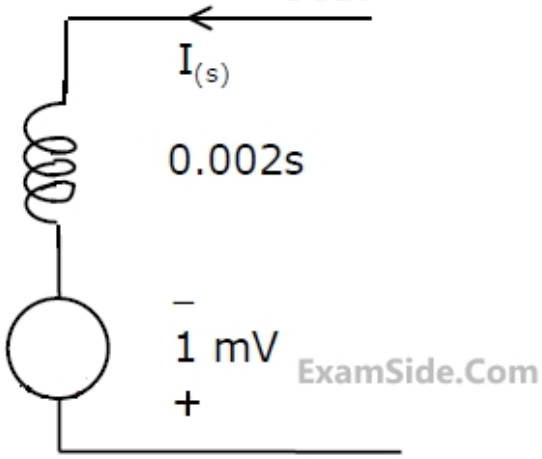
(C) 0.50 exp(-12.5t) mA

(D) 0.25 exp (-6.25t) mA

Q36. GATE ECE 2006

Open solution

A 2mH inductor with some initial current can be represented as shown below, where s is the Laplace Transform variable. The value of initial current is:



(A) 0.5 A

(B) 2.0 A

(C) 1.0 A

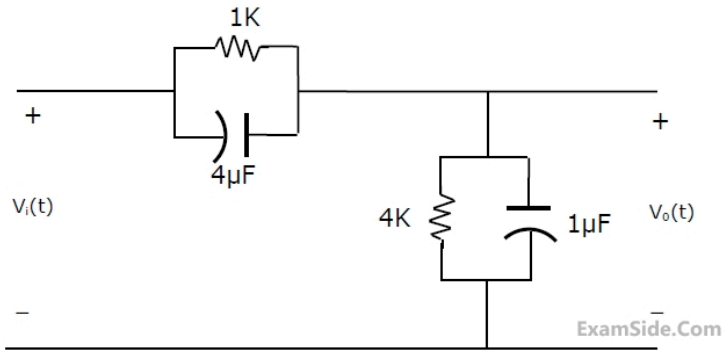
(D) 0.0 A

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Q37. GATE ECE 2006

Open solution

In the figure shown below, assume that all the capacitors are initially uncharged. If $v_i(t) = 10\ u(t)$ Volts, $v_0(t)$ is given by



(A) $8\ e^{-0.004t}$ Volts

(B) $8\ (1 - e^{-0.004t})$ Volts

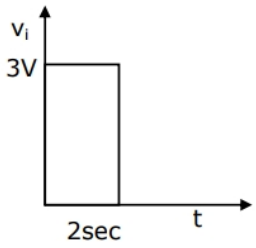
(C) 8 u(t) Volts

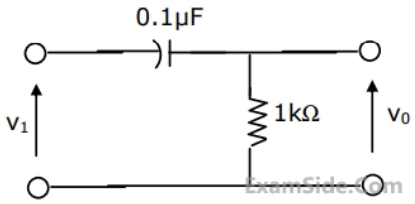
(D) 8 Volts

Q38. GATE ECE 2005

Open solution

A square pulse of 3 volts amplitude is applied to C-R circuit shown in figure. The capacitor is initially uncharged. The output voltage v_0 at time t=2 sec is





(A) 3 V

(B) -3 V

(C) 4 V

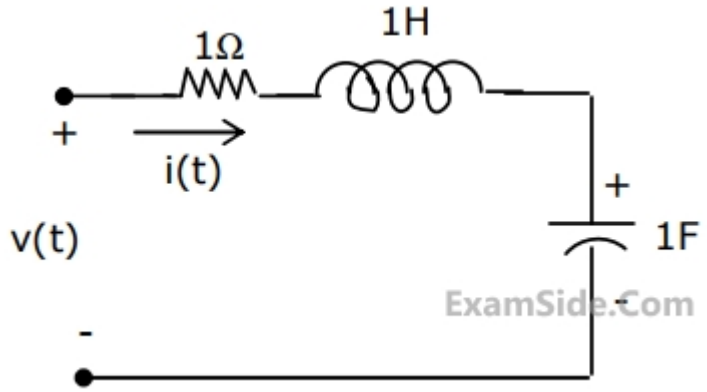
(D) -4 V

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Q39. GATE ECE 2004

Open solution

The circuit shown in Fig, has initial current $i_L(0^-) = 1\ A$ through the inductor and an initial voltage $v_C(0^-) = -1\ V$ across the capacitor. For input v(t) = u(t), the Laplace transform of the current i(t) for t ≥ 0 is

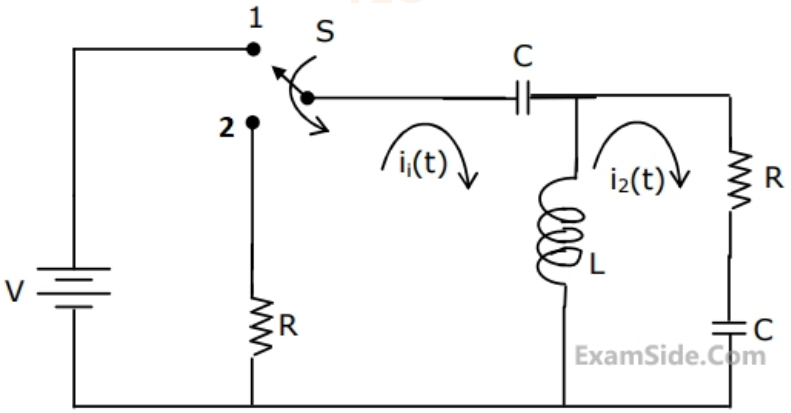


(A) $\frac{s}{s^2+s+1}$ (B) $\frac{s+2}{s^2+s+1}$ (C) $\frac{s-1}{s^2+s+1}$ (D) $\frac{s+1}{s^2+s+1}$

Q40. GATE ECE 2003

Open solution

The circuit is given in figure.Assume that the switch S is in position 1 for a long time and thrown to position 2 at t = 0.



At t = 0^+, the current i_1 is

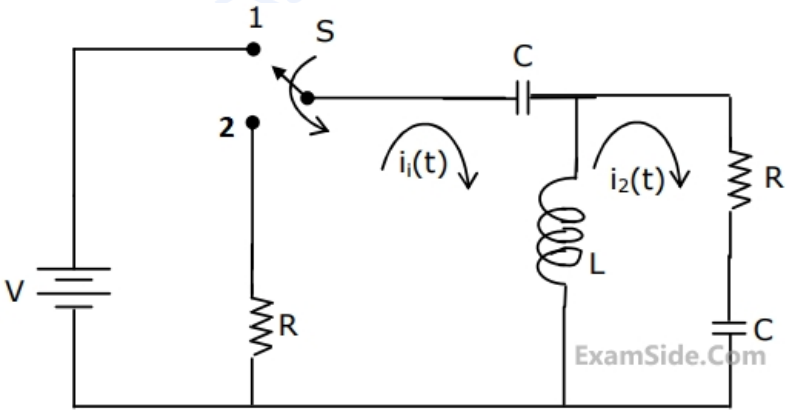
(A) $\frac{-V}{2R}$ (B) $\frac{-V}{R}$ (C) $\frac{-V}{4R}$ (D) zero

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Q41. GATE ECE 2003

Open solution

The circuit is given in figure.Assume that the switch S is in position 1 for a long time and thrown to position 2 at t = 0.



$I_1(s)$ and $I_2(s)$ are the Laplace transforms of $i_1(t)$ and $i_2(t)$ respectively. The equations for the loop currents $I_1(s)$ and $I_2(s)$ for the circuit shown in figure, after the switch is brought from position 1 to position 2 at t = 0, are

(A) $$\begin{bmatrix} R+Ls+\frac{1}{Cs} & -Ls \\ -Ls & R+\frac{1}{Cs} \end{bmatrix}\begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix}=\begin{bmatrix} \frac{V}{s} \\ 0 \end{bmatrix}$$

(B) $$\begin{bmatrix} R+Ls+\frac{1}{Cs} & -Ls \\ -Ls & R+\frac{1}{Cs} \end{bmatrix}\begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix}=\begin{bmatrix} -\frac{V}{s} \\ 0 \end{bmatrix}$$

(C) $$\begin{bmatrix} R+Ls+\frac{1}{Cs} & -Ls \\ -Ls & R+Ls+\frac{1}{Cs} \end{bmatrix}\begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix}=\begin{bmatrix} \frac{V}{s} \\ 0 \end{bmatrix}$$

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(D)

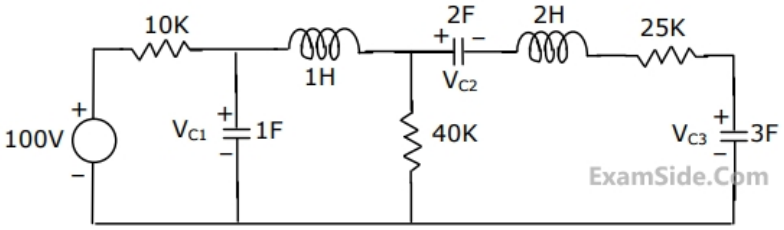
$$\begin{bmatrix} R+Ls+\frac{1}{Cs} & -Ls \\ -Ls & R+Ls+\frac{1}{Cs} \end{bmatrix} \begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} -\frac{V}{s} \\ 0 \end{bmatrix}$$

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Q42. GATE ECE 1996

Open solution

The voltage V_{C1}, V_{C2} and V_{C3} across the capacitors in the circuit in Fig., under steady state, are respectively

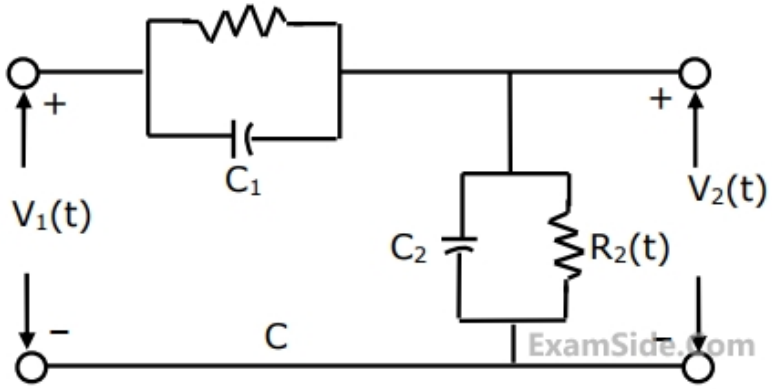


(A) 80 V, 32 V, 48 V (B) 80 V, 48 V, 32 V (C) 20 V, 8V, 12 V (D) 20 V, 12 V, 8 V

Q43. GATE ECE 1992

Open solution

For the compensated attenuator of figure, the impulse response under the condition $R_1C_1 = R_2C_2$ is:



(A) $\frac{R_2}{R_1+R_2}\left[1 - e^{\frac{1}{R_1C_1}}\right]u(t)$

(B) $\frac{R_2}{R_1+R_2}\delta(t)$

(C) $\frac{R_2}{R_1+R_2}u(t)$

(D) $\frac{R_2}{R_1+R_2}e^{\frac{1}{R_1C_1}}u(t)$

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Q44. GATE ECE 1990

If the Laplace transform of the voltage across a capacitor of value of $\frac{1}{2}$ F is $V_C(s) = \frac{s+1}{s^3+s^2+s+1}$, the value of the current through the capacitor at t = 0^+ is

(A) 0 A (B) 2 A (C) $\frac{1}{2}$ A (D) 1 A

Open solution

Q45. GATE ECE 1989

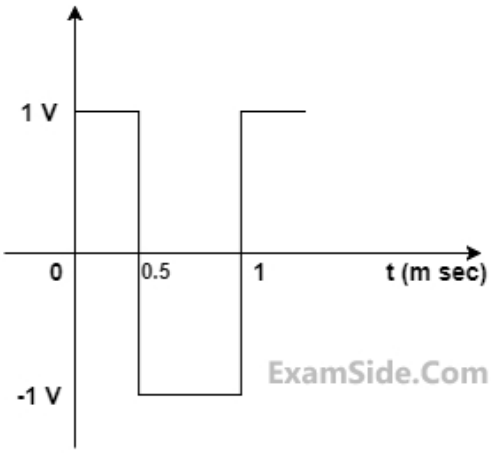
A 10 Ω resistor, a 1 H inductor and 1 μF capacitor are connected in parallel.The combination is driven by a unit step current.Under the steady state condition, the source current flows through

(A) the resistor
(B) the inductor
(C) the capacitor only
(D) all the three elements

Open solution

Q46. GATE ECE 1987

A square waveform as shown in the figure is applied across 1 mH ideal inductor.The current through the inductor is a triangular wave of ..... peak amplitude.



(A) triangular wave with peak amplitude of 0.5 Amp.
(B) triangular wave with peak amplitude of 1.0 Amp.
(C) triangular wave with peak amplitude of 2.0 Amp.
(D) sine wave with peak amplitude of 0.5 Amp.

Open solution

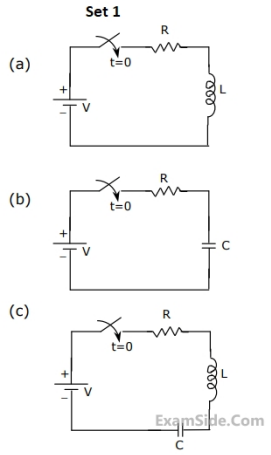
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Q47. GATE ECE 1996

Open solution

In the circuit shown in Fig (a) - (c), assuming initial voltages across capacitors and current through the inductors to be zero at the time of switching (t=0), then at any time t>0.
Match each of the items of **Set 1**, with the appropriate item of the **Set 2**.



Set 2

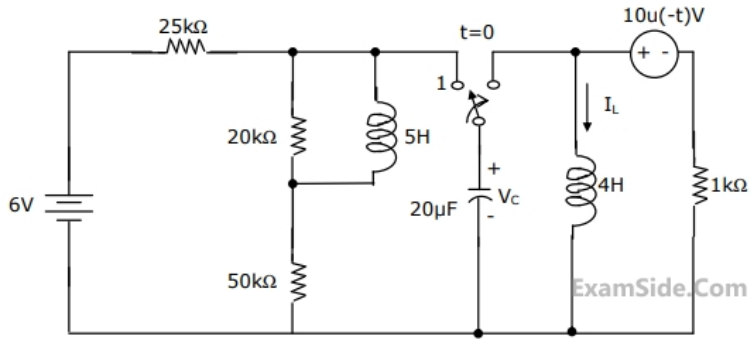
(1) current increases monotonically with time
(2) current decreases monotonically with time
(3) current remains constant at V/R
(4) current first increases, then decreases
(5) no current can ever flow

Marks 5 7 questions

Q48. GATE ECE 2002

Open solution

The switch in Fig. has been in position 1 for a long time and is then moved to position 2 at $t = 0$.
(a) Determine $V_C\,(0^+)$ and $I_L\,(0^+)$
(b) Determine $\frac{dV_C(t)}{dt}$ at $t = 0^+$
(c) Determine $V_C\,(t)$ for $t > 0$



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Q49. GATE ECE 2001

Open solution

The circuit shown in Fig. is operating in steady-state with switch S_1 closed. The switch S_1 is opened at $t = 0$.

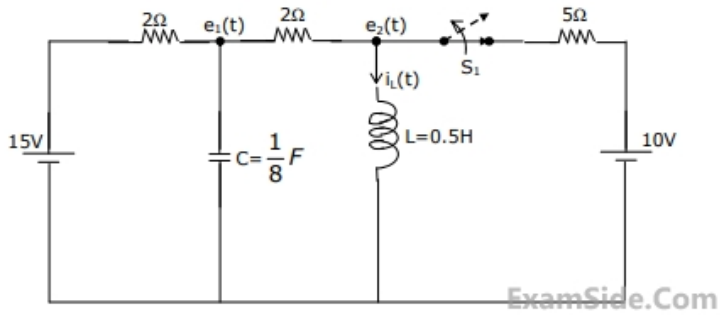


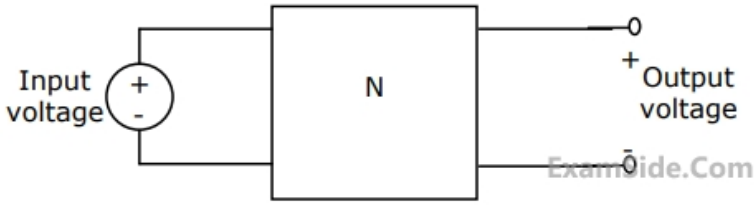
Fig.

(a) Find $i_L\,(0^+)$.
(b) Find $e_1\,(0^+)$.
(c) Using nodal equations and Laplace transform approach, find an expression for the voltage across the capacitor for all $t = 0$.

Q50. GATE ECE 2000

Open solution

The network N in Fig. consists only of two elements: a resistor of 1Ω and an inductor of L Henry. A $5\ V$ source is connected at the input at $t = 0$ seconds. The inductor current is zero at $t = 0$. The output voltage is found to be $5e^{-3t}\ V$, for $t = 0$.



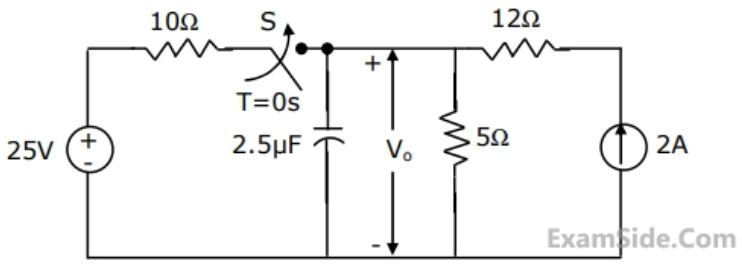
(a) Find the voltage transfer function of the network.
(b) Find L, and draw the configuration of the network.
(c) Find the impulse response of the network.

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Q51. GATE ECE 1999

Open solution

In the circuit of figure, the switch 'S' has remained open for a long time. The switch closes instantaneously at $t = 0$.



n

(a) Find V_0 for $t \leq 0$ and as $t \to \infty$.

(b) Write an expression for V_0 as a function of time for $0 \leq t \leq \infty$.

(c) Evaluate V_0 at $t = 25$ μsec.

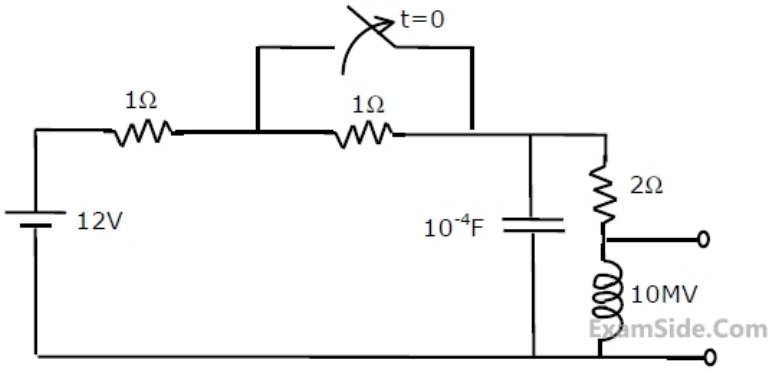
Q52. GATE ECE 1994

Open solution

The circuit shown in the figure, is initially in its steady state. Switch is opened at t = 0.

(1) Determine the initial voltage, $V_C(0^-)$, across the capacitor, and the initial current, $i_L(0^-)$, through the indicator.

(2) Calculate the voltage, $V_L(t)$, across the inductance for t > 0.

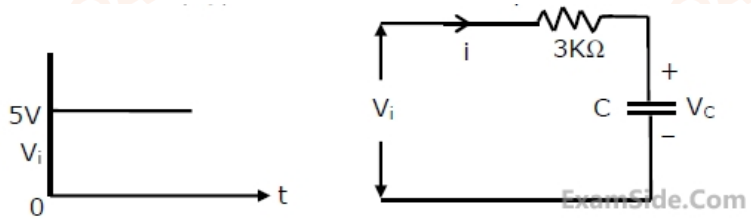


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Q53. GATE ECE 1993

In the following circuit the capacitance varies as C = KQ, where K is a constant equal to 0.5 Farads/Coulomb and Q, the charge on the capacitor in Coulombs. Determine the current through the circuit and sketch the voltage waveform across the capacitor (V_C) for a step input V_i as shown in figure.

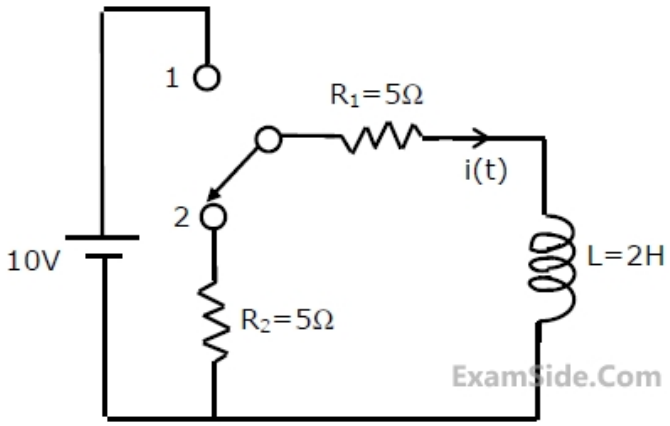
Open solution



Q54. GATE ECE 1991

The network shown in figure is initially under steady-state condition with the switch in position 1. The switch is moved from position 1 to position 2 at t = 0. Calculate the current i(t) through R_1 after switching.

Open solution



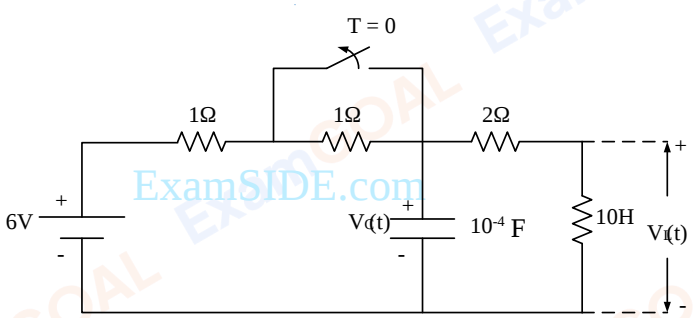
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Q55. GATE ECE 1988

The circuit shown below is initially in a steady state condition. At $t = 0$, the switch 'S' is opened.
(i) Determine the initial voltage $V_c\,(0^-)$ across the capacitor and initial current $i_L\,(0^-)$ through the inductor.
(ii) Find the voltage $v_L\,(t)$ across the inductor for $t > 0$.

Open solution



Marks 10 1 questions

Q56. GATE ECE 1990

The switch S in figure is thrown to position 'a' at $t = 0$. At $t = 1$ second, the switch is thrown to position 'b'. Find the current through R_2 for $t > 1$ sec.

Open solution

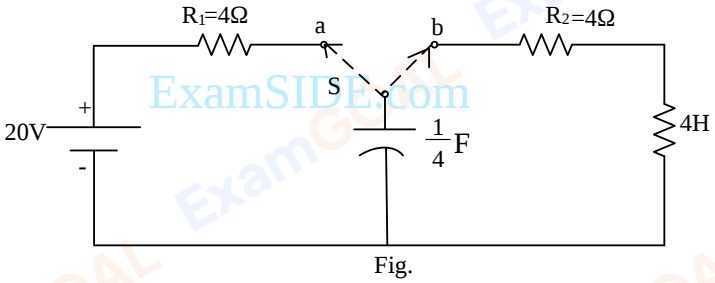


Fig.

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Answer Key

Marks 1

1. **0.75**	2. **0.250**	3. **4**	4. **1.42 to 1.45**	5. **D**
6. **D**	7. **C**	8. **1.25**	9. **A**	10. **D**
11. **A**	12. **B**	13. **C**	14. **C**	15. **C**
16. **B**	17. **C**			

Marks 2

18. **B**	19. **0.13 to 0.14**	20. **0.284 to 0.348**	21. **8.16**	22. **100**
23. **-1**	24. **1.5**	25. **2.528**	26. **31.25**	27. **A**
28. **A**	29. **A**	30. **B**	31. **A**	32. **D**
33. **B**	34. **C**	35. **A**	36. **A**	37. **C**
38. **B**	39. **B**	40. **A**	41. **D**	42. **B**
43. **B**	44. **C**	45. **B**	46. **A**	

Marks 3

47. **(a-1),(b-2),(c-4)**	

Marks 5

48. **(a)** $V_C(0^+) = 4V,\ I_L(0^+) = 0A$ **(b)** $-700\,V/\sec$ **(c)** $V_C(t) = 4\left[e^{-25t}\cos(109t) - 1.38e^{-25t}\sin(109t)\right]V$	49. **(a)** $6\,A$ **(b)** $4V$ **(c)** $8u(t) - 4e^{-4t}\sin(4t)ut$	50. **(a)** $\frac{s}{s+3}$ **(b)** $\frac{1}{3}H$ **(c)** $1\delta(t) - 3e^{-3t}u(t)$	51. **(a)** $10V$ **and** $15V$ **(b)** $15 - 5e^{-0.12t}$ **(c)** $14.75V$	52. **(a) 8 V, 4 A** **(b) 0 V**
53. **Question is incomplete**	54. $i(t) = 2e^{-5t}u(t)$			

Marks 8

55. **(a)** $4\,V,\quad 2\,A$	

Marks 10

56. **Current** $= \left[3.637\,e^{-0.5(t-1)}\sin\left(\frac{\sqrt{3}}{2}(t-1)\right)u(t-1)\right]$ A	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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